

Mamba-3: Improved Sequence Modeling using State Space Principles

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ONE-LINE TAKEAWAY

At the **1.5B** scale, Mamba-3 improves average downstream accuracy by **0.6** percentage points compared to the next best model (Gated DeltaNet), with Mamba-3's MIMO variant further improving accuracy by another **1.2** points for a total **1.8** point gain.

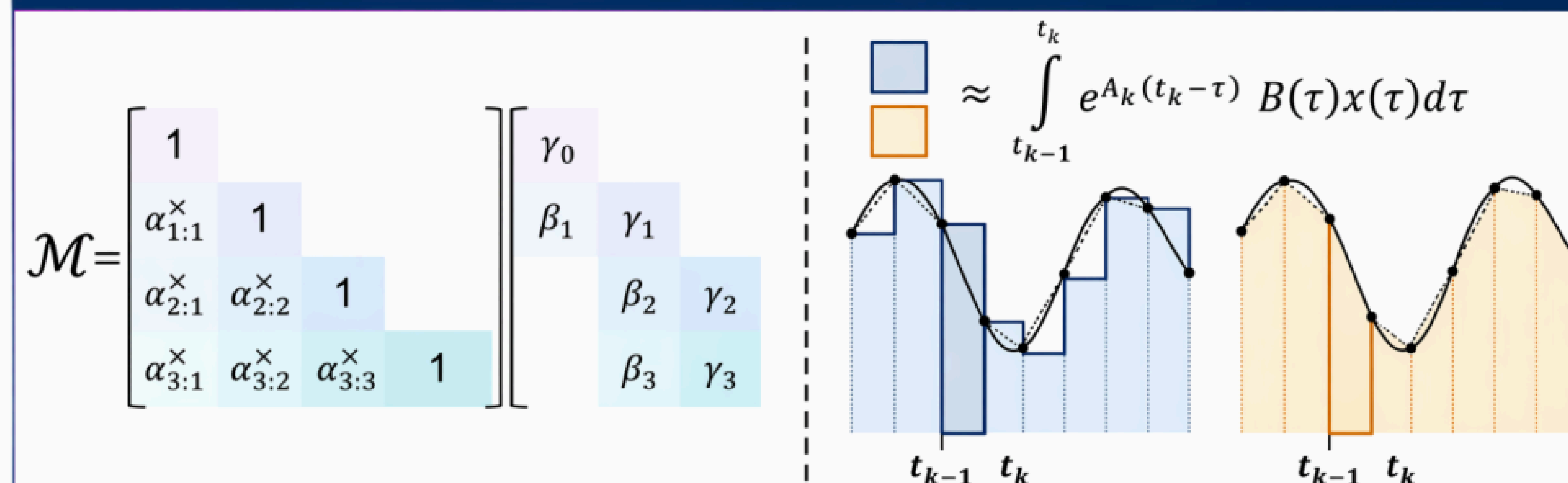
PROBLEM

- Transformer-based models have quadratic compute and linear memory requirements, making inference expensive.
- Many recent linear models trade off model quality and capability for algorithmic efficiency, failing on tasks such as state tracking.

METHOD

- Exponential-Trapezoidal Discretization:** a more expressive recurrence derived from SSM discretization.
- Complex-valued State Update Rule:** enables richer state tracking.
- Multi-Input, Multi-Output (MIMO) Formulation:** improves model performance without increasing decode latency.
- Architectural refinements** combining these components.

Figure 1: Left: The structured mask induced by the exponential-trapezoidal rule

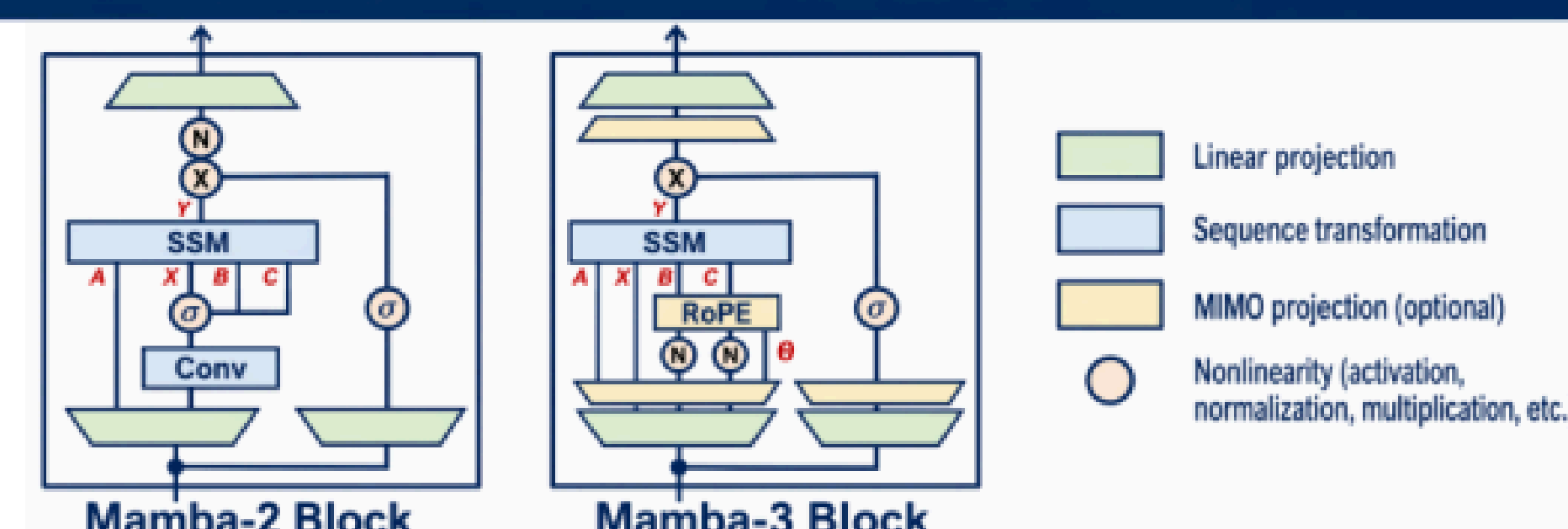


is a 1-semiseparable matrix composed with a 2-band matrix.⁴

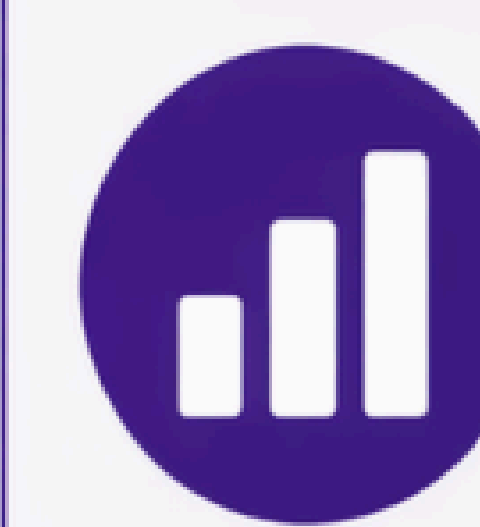
is a 1-semiseparable matrix composed with a 2-band matrix:⁴

$$L = \begin{bmatrix} \gamma_0 & & & \\ (\gamma_0 \alpha_1 + \beta_1) & \gamma_1 & & \\ \alpha_2 (\gamma_0 \alpha_1 + \beta_1) & (\gamma_1 \alpha_2 + \beta_2) & \gamma_2 & \\ \vdots & \vdots & \ddots & \\ \alpha_{T-2} (\gamma_0 \alpha_1 + \beta_1) & \cdots & \gamma_{T-1} & \end{bmatrix} = \begin{bmatrix} 1 & & & \\ \alpha_1 & 1 & & \\ \alpha_2 \alpha_1 & \alpha_2 & 1 & \\ \vdots & \vdots & \ddots & \\ \alpha_{T-1} & \cdots & 1 & \end{bmatrix} \begin{bmatrix} \gamma_0 & & & \\ \beta_1 & \gamma_1 & & \\ 0 & \beta_2 & \gamma_2 & \\ \vdots & \vdots & \ddots & \\ 0 & \cdots & \gamma_{T-1} & \end{bmatrix}. \quad (7)$$

Figure 2: Contrasting Mamba-2 and Mamba-3 Architectures: Key updates include exp



RESULTS



At 1.5B scale, Mamba-3 (MIMO) improves downstream language modeling accuracy by

+2.2
over
Transformers

+1.9
over
Mamba-2

+1.8
over
GDN



Across state size experiments, Mamba-3 (MIMO) with state size **64** matches the perplexity of Mamba-2 with state size **128**.



Mamba-3 (SISO) improves over GDN by **+0.6** points.



Mamba-3 (MIMO) increases decoding FLOPs by up to **4x** relative to Mamba-2 at fixed state size while maintaining similar wall-clock decode latency.

KEY NUMBERS

1.5B

model scale used in evaluation.

0.6

improvement in accuracy over Gated DeltaNet (Mamba-3 SISO).

1.8

total accuracy gain over GDN (Mamba-3 MIMO).

64

state size used in Mamba-3 MIMO achieving same perplexity as Mamba-2 at 128.